

Certified Penetration Testing Professional

Methodology: Wireless Penetration Testing

Penetration Tester:			
Organization:			
Date:		**Location:**	

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Test 1: Wireless Local Area Network (WLAN) Penetration Testing

Test 1.1: Discover the Wireless Networks

Target Organization	
URL	
Techniques Used	1. 2. 3. 4. 5.
Access Points Discovered	1. 2. 3. 4. 5.
Soft Access Points Discovered	1. 2. 3. 4. 5.

Data of Discovered Access Points				
Access Points Discovered	**SSID**	**BSSID**	**Encryption Technique**	**Beacon Strength**
1.				
2.				
3.				

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4.				
5.				
6.				

Tools/Services Used	1. 2. 3. 4. 5.

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Test 1.2: Check Physical Security of AP

Target Organization			
URL			
Physical Location of Authorized APs			
Physical Access to APs Is Controlled?		☐ Yes	☐ No
Network's Physical Security Policy			
Details of Authorized Persons having Physical Access to APs	1. 2. 3. 4. 5.		

Details of Access Points				
APs Discovered	**Location**	**Model Type**	**MAC Address**	**IP Address**
1.				
2.				
3.				
4.				
5.				

Tools/Services Used	1. 2. 3. 4. 5.

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Test 1.3: Detect Wireless Connections

Target Organization			
URL			
Scanning Methodologies			
Wireless Connections Detected using Active Scanning		**Wireless Connections Detected using Passive Scanning**	
1.		1.	
2.		2.	
3.		3.	
4.		4.	
5.		5.	

Gathered AP Information					
Access Points Discovered		**SSID**	**BSSID**	**Encryption Technique**	**Beacon Strength**
1.					
2.					
3.					
4.					
5.					
6.					
7.					
8					
9.					
10.					

Identified Network Vulnerabilities	1.
	2.
	3.
	4.
	5.

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Tools/Services Used	1. 2. 3. 4. 5.

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Test 1.4: Sniff Traffic between the AP and Linked Devices

Target Organization	
URL	
Information gathered from Sniffed Traffic	
BSSID	STATION
PWR	PWR
Beacons	Packets
#Data	Probes
CH	Others:
HB	
ENC	
ESSID	
BSSID	
Sensitive Information obtained Through Sniffing	1. 2. 3. 4. 5. 6. 7.
Tools/Services Used	1. 2. 3. 4. 5.

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Test 1.5: Create a Rogue Access Point and Try to Create a Promiscuous Client

Target Organization		
URL		
Details of Rogue Access Point		
Location of Rogue Access Point		
SSID Broadcast Disabled?	☐ Yes	☐ No
Details of Firewall Used to the Place the Rogue AP		
Promiscuous Client Creation Successful?	☐ Yes	☐ No
Details of Users Connected to Rogue AP	1. 2. 3. 4. 5.	
Sensitive Information Gathered from Network Data	1. 2. 3. 4. 5.	
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 1.6: Use a Wireless Honeypot to Discover Vulnerable Wireless Clients

Target Organization			
URL			
Details of Honeypot or Evil Twin AP Used			
Discovered Vulnerable Wireless Clients?		☐ Yes	☐ No
Details of Vulnerable Wireless Clients	1. 2. 3. 4. 5.		
Captured any Email or FTP Connections?		☐ Yes	☐ No
Details of Email or FTP Connections Captured	1. 2. 3. 4. 5.		
Able to Access the User's File Shares?		☐ Yes	☐ No
Details of User's File Shares Captured	1. 2. 3. 4. 5.		
Captured Login Credentials via Captive Portal or Spoofed DNS Caching?		☐ Yes	☐ No

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List of User Credentials Captured	1. 2. 3. 4. 5.
Tools/Services Used	1. 2. 3. 4. 5.

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Test 1.7: Perform a Denial-of-Service Attack (De-authentication Attack)

Target Organization		
URL		
Commands Used		
Is De-authentication Attack Successful?	☐ Yes	☐ No
Response Received		
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 1.8: Attempt Rapid Traffic Generation

Target Organization	
URL	
Commands Used	
Captured Source MAC Addresses	1. 2. 3. 4. 5.
Captured Destination MAC Addresses	1. 2. 3. 4. 5.
Discovered Hosts on the Wireless Network	1. 2. 3. 4. 5.
Discovered Hosts on the Bridged or Wired LAN	1. 2. 3. 4. 5.

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Is Replay Attack on the Target Wireless Network Successful?	☐ Yes ☐ No
WEP Keys Cracked	1. _____ 2. _____ 3. _____ 4. _____ 5. _____
Tools/Services Used	1. _____ 2. _____ 3. _____ 4. _____ 5. _____

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Test 1.9: Attempt Single-packet Decryption

Target Organization	
URL	
Commands Used	
Techniques Used	
Source MAC Address	
Destination MAC Address	
Recovered Plain Text	
Security Level of the Target Network	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 1.10: Perform an ARP Poisoning Attack

Target Organization		
URL		
Fake MAC Address Used		
IP Address of the AP		
Is ARP Poisoning Attack Successful?	☐ Yes	☐ No
Details Hosts/Subnet Compromised	1. 2. 3. 4. 5.	
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 1.11: Try to Inject the Encrypted Packet

Target Organization			
URL			
Commands Used			
Is Encrypted Packet Injection Attack Successful?		☐ Yes	☐ No
List of APs Responded to Broadcast Probes	1. 2. 3. 4. 5.		
Identified Network Vulnerabilities	1. 2. 3. 4. 5.		
Ping Response Time to the AP			
Hidden SSIDs Detected	1. 2. 3. 4. 5.		
Tools/Services Used	1. 2.		

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	3. _____ 4. _____ 5. _____

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Test 1.12: Crack WPA-PSK Keys

Target Organization	
URL	
Command Used to Monitor Traffic	
Command Used to Collect Traffic Data	

Information gathered by Cracking WPA-PSK Keys	
BSSID:	CIPHER:
PWR:	AUTH:
RXQ:	ESSID:
Beacons:	Others:
#Data:	
CH:	
MB:	
ENC:	

Information Extracted from WPA2 4-way handshake communication	1. 2. 3. 4. 5.
Techniques Used to Recover the Keys	

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Recovered WPA-PSK Keys	
Tools/Services Used	1. 2. 3. 4. 5.

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Test 1.13: Crack WPA/WPA2 Enterprise Mode

Target Organization		
URL		
Commands Used		
Details of Fake AP Used		
IS the Man-in-the-Middle (MITM) Attack Successful?	☐ Yes	☐ No
Captured and Recovered Authentication Credentials?	☐ Yes	☐ No
Recovered Authentication Credentials		
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 1.14: Check for MAC Filtering

Target Organization		
URL		
Commands Used		
Is Target Access Point Using MAC Filtering?	☐ Yes	☐ No
Fake Auth Commands		
Is Authentication Successful?	☐ Yes	☐ No
Is Association Successful?	☐ Yes	☐ No
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 1.15: Spoof the MAC Address

Target Organization		
URL		
Commands Used		
Details of Legitimate Device Used for Impersonation		
Name of the SSID tested		
Spoofed MAC Address	1. 2. 3. 4. 5.	
New MAC Address and Vendor Settings		
Is MAC Spoofing Successful?	☐ Yes	☐ No
Identified Attack Vectors	1. 2. 3. 4. 5.	
Tools/Services Used	1. 2. 3.	

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Test 1.16: Create a Direct Connection to the Wireless Access Point

Target Organization		
URL		
DHCP Enabled		
Wireless AP	☐ Yes	☐ No
Laptop	☐ Yes	☐ No
Does the Target Wireless Network is Allowing Direct Connections to Malicious Devices?	☐ Yes	☐ No
IP Address of Wireless AP		
Tools/Services Used	1. 2. 3. 4. 5.	

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Test 1.17: Additional Wireless Penetration Testing Tools

Target Organization	
URL	
Commands Used	

Wireless Adapter	BSSID	Associated Vulnerabilities
1.	1.	1.
2.	2.	2.
3.	3.	3.
4.	4.	4.

Information Gathered from Wireless Penetration Testing	1. 2. 3. 4. 5.
Tools/Services Used	1. 2. 3. 4. 5.

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Test 2: RFID Penetration Testing

Test 2.1: Perform Reverse Engineering

Target Organization		
URL		
Target RFID System		
Methods/Techniques Used	1. 2. 3. 4. 5.	
Is RFID Reverse Engineering Successful?	☐ Yes	☐ No
Step-1: Visual inspection		
Manufacturer Details		
RFID Model		
RFID Standard		
Other Information, if any		
Step-2: Monitoring Coupling and Frequencies		
RFID Operational Frequency		
Suspected Frequencies Recorded using Spectrum Analyzer/Oscilloscope		
Step-3: Monitoring Energy supply and modulations		
How the RFID Tag Powered	☐ Own Battery	☐ Reader Signal

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Details of RFID Battery	
Modulation Parameters	
Step-4: Line encoding, Syntax inference, and Protocol inference using signal Spectrogram	
Symbol Coding Implemented in the RFID Tag	
Step-5: Cryptanalysis	
Encoding Techniques Used	
Other Information Collected	1. 2. 3. 4. 5.
Identified Vulnerabilities in RFID Systems	1. 2. 3. 4. 5.
Tools/Services Used	1. 2. 3. 4. 5.

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Test 2.2: Perform Power Analysis Attack

Target Organization			
URL			
Target RFID System			
Details of Devices Used in Power Analysis	1. 2. 3. 4. 5.		
Performed Power Analysis Attack Successfully?		☐ Yes	☐ No
Did the Device Leaked any Information During Cryptographic Operations?		☐ Yes	☐ No
Identified Secret Keys			
Devices/Tools Used	1. 2. 3. 4. 5.		

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Test 2.3: Perform Eavesdropping

Target Organization		
URL		
Target RFID System		
Equipment Used for Eavesdropping	1. 2. 3. 4. 5.	
Eavesdropping the Legitimate Transmission between the RFID Tag and the Reader Successful?	☐ Yes	☐ No
Sensitive Information Obtained through Eavesdropping	1. 2. 3. 4. 5.	
Identified Vulnerabilities	1. 2. 3. 4. 5.	
Devices/Tools Used	1. 2. 3. 4. 5.	

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Test 2.4: Perform an MITM Attack

Target Organization			
URL			
Target RFID System			
Equipment Used for Eavesdropping and Intercepting Traffic	1. 2. 3. 4. 5.		
Interception of the Communication between the RFID Tag and the Reader Successful?		☐ Yes	☐ No
Data Transmitted in Clear Text?		☐ Yes	☐ No
Information Recovered	1. 2. 3. 4. 5. 6. 7. 8. 9.		
Devices/Tools Used	1. 2. 3. 4. 5.		

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Test 2.5: Perform a DoS Attack

Target Organization		
URL		
Target RFID System		
Performed DoS Attack Successfully against:		
1. RFID Tag **2. RFID Reader** **3. Backend Server**	☐ YES ☐ YES ☐ YES	☐ NO ☐ NO ☐ NO
Techniques Used to Perform DoS Attack	1. 2. 3. 4. 5.	
Response Received	1. 2. 3. 4. 5.	
Devices/Tools Used	1. 2. 3. 4. 5.	

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Test 2.6: Perform RFID Cloning/Spoofing

Target Organization		
URL		
Target RFID System		
Captured Data from the Legitimate RFID Tag and Created a Clone of it using a New Chip Successfully?	☐ Yes	☐ No
Overwritten Existing RFID Tag Data with the Spoofed Data (Obtained by Eavesdropping) Successfully?	☐ Yes	☐ No
Data Transmitted from an RFID Tag		
Devices/Tools Used	1. 2. 3. 4. 5.	

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Test 2.7: Perform an RFID Replay Attack

Target Organization		
URL		
Target RFID System		
Techniques Used	1. 2. 3. 4. 5.	
Intercepted Communication between the RFID Reader and Tag, and Captured an RFID Signal Successfully?	☐ Yes	☐ No
Is the RFID Replay Attack Successful?	☐ Yes	☐ No
Information Gathered	1. 2. 3. 4. 5.	
Identified Vulnerabilities	1. 2. 3. 4. 5.	
Devices/Tools Used	1. 2. 3. 4.	

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Test 2.8: Perform a Virus Attack

Target Organization			
URL			
Target RFID System			
Injected Infective Viruses to the Memory Space of RFID Tags Successfully?		☐ Yes	☐ No
Compromised Backend RFID Middleware Systems via an SQL Injection Attack?		☐ Yes	☐ No
Identified Vulnerabilities that leads to Virus Attacks	1. 2. 3. 4. 5.		
Devices/Tools Used	1. 2. 3. 4. 5.		

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Test 3: NFC Penetration Testing

Test 3.1: Perform Eavesdropping

Target Organization		
URL		
Target NFC Device		
Eavesdropping of the Communication between NFC Devices Successful?	☐ Yes	☐ No
Information Obtained	1. 2. 3. 4. 5.	
Devices/Tools Used	1. 2. 3. 4. 5.	

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Test 3.2: Perform a Data Modification Attack

Target Organization		
URL		
Target NFC Device		
Interfered with the NFC Data Exchange Successfully?	☐ Yes	☐ No
Is Modifying the Control-Flow of Programs Successful?	☐ Yes	☐ No
Running Data Captured	1. 2. 3. 4. 5.	
Memory Corruption Techniques Used to Override Data	1. 2. 3. 4. 5.	
Identified Vulnerabilities	1. 2. 3. 4. 5.	
Devices/Tools Used	1. 2. 3. 4. 5.	

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Test 3.3: Perform Data Corruption Attack

Target Organization		
URL		
Target NFC Device		
Performed Data Corruption Attack Successfully?	☐ Yes	☐ No
Techniques Used to Perform Data Corruption	1. 2. 3. 4. 5.	
Identified Vulnerabilities	1. 2. 3. 4. 5. 6. 7. 8. 9. 10.	
Devices/Tools Used	1. 2. 3. 4. 5.	

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Test 3.4: Perform a MITM Attack

Target Organization			
URL			
Target NFC Device			
Eavesdropped, Manipulated and Transmitted the Data to the NFC Reader Successfully?		☐ Yes	☐ No
Identified Vulnerabilities	1. 2. 3. 4. 5.		
Devices/Tools Used	1. 2. 3. 4. 5.		

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